

MH2500 Probability and Introduction to Statistics

Slides for Week 10-2

Topics: Covariance

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Covariance – Definition

- The variance of a random variable measures its variability.
- The **covariance** of two random variables measures their joint variability, or **degree of association**.

Definition

If X and Y are jointly distributed random variables with expectations μ_X and μ_Y , respectively, then **the covariance of X and Y** is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)],$$

provided that the expectation exists.

Covariance is the average value of the product of the deviation of X from its mean and the deviation of Y from its mean.

- Positive covariance means both X and Y tend to be larger than their respective means or both X and Y tend to be smaller than their respective means.
- Negative covariance means X tends to be larger (smaller) than its mean while Y tends to be smaller (respectively larger) than its mean.

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Proof:

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mu_Y - \mathbb{E}(Y)\mu_X + \mu_X\mu_Y \\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).\end{aligned}$$

So $\text{Cov}(X, X) = \text{Var}(X)$.

- If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X) \mathbb{E}(Y)$ and $\text{Cov}(X, Y) = 0$.
- The converse is not true.

$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent.}$

Example

Let X be uniform on $\{-1, 0, 1\}$ and $Y = \mathbf{1}_{\{X=0\}}$.

- It is always true that $XY = 0$, and thus $\mathbb{E}(XY) = 0$.
- $\mathbb{E}(X) = 0$ hence $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X) \mathbb{E}(Y) = 0$ but X and Y are not independent.

Covariance of linear combinations

- Recall that $\mathbb{E}(a) = a$. Hence

$$\begin{aligned}\text{Cov}(a, Y) &= \mathbb{E}[(a - \mathbb{E} a)(Y - \mathbb{E} Y)] \\ &= \mathbb{E}[0 \cdot (Y - \mathbb{E} Y)] \\ &= 0.\end{aligned}$$

- Recall $\mathbb{E}(aX) = a\mathbb{E}(X)$. Hence

$$\begin{aligned}\text{Cov}(aX, Y) &= \mathbb{E}[(aX - \mathbb{E}(aX))(Y - \mathbb{E} Y)] \\ &= \mathbb{E}[(aX - a\mathbb{E} X)(Y - \mathbb{E} Y)] \\ &= a\mathbb{E}[(X - \mathbb{E} X)(Y - \mathbb{E} Y)] \\ &= a\text{Cov}(X, Y).\end{aligned}$$

$$\begin{aligned}
\text{Cov}(X + Y, Z) &= \mathbb{E}[(X + Y - \mathbb{E}(X + Y))(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X + Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X)(Z - \mathbb{E} Z) + (Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= \mathbb{E}[(X - \mathbb{E} X)(Z - \mathbb{E} Z)] + \mathbb{E}[(Y - \mathbb{E} Y)(Z - \mathbb{E} Z)] \\
&= \text{Cov}(X, Z) + \text{Cov}(Y, Z).
\end{aligned}$$

Therefore $\text{Cov}(\cdot, \cdot)$ is linear w.r.t. the first argument:

$$\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z).$$

and same w.r.t. the second argument since $\text{Cov}(\cdot, \cdot)$ is symmetric.

Combine the above, $\text{Cov}(aW + bX, cY + dZ)$

$$\begin{aligned}
&= \text{Cov}(aW, cY) + \text{Cov}(bX, cY) + \text{Cov}(aW, dZ) + \text{Cov}(bX, dZ) \\
&= ac\text{Cov}(W, Y) + bc\text{Cov}(X, Y) + ad\text{Cov}(W, Z) + bd\text{Cov}(X, Z).
\end{aligned}$$

Theorem

Suppose that $U = a + \sum_{i=1}^n b_i X_i$ and $V = c + \sum_{j=1}^m d_j Y_j$. Then

$$\text{Cov}(U, V) = \sum_{i=1}^n \sum_{j=1}^m b_i d_j \text{Cov}(X_i, Y_j).$$

Since $\text{Var}(X) = \text{Cov}(X, X)$,

$$\begin{aligned}\text{Var}(X + Y) &= \text{Cov}(X + Y, X + Y) \\ &= \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(Y, X) + \text{Cov}(Y, Y) \\ &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).\end{aligned}$$

Corollary 1

$$\text{Var} \left(a + \sum_{i=1}^n b_i X_i \right) = \sum_{i=1}^n \sum_{j=1}^n b_i b_j \text{Cov}(X_i, X_j).$$

Corollary 2

Suppose the X_i 's are independent. Then

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i).$$

Example: Sample Variance

Let $X_1, \dots, X_n$ be i.i.d. random variables having expected value μ and variance σ^2 , and let $\bar{X} = (\sum_i X_i)/n$ be the sample mean.

The **sample variance** is defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$\mathbb{E}(S^2) = \frac{1}{n-1} \sum_{i=1}^n \mathbb{E}[(X_i - \bar{X})^2] = \sigma^2.$$

Proof:

$$\begin{aligned} \mathbb{E}[(X_i - \bar{X})^2] &= \mathbb{E}((X_i - \mu + \mu - \bar{X}))^2 \\ &= \mathbb{E}\left((X_i - \mu)^2 + 2(X_i - \mu)(\mu - \bar{X}) + (\mu - \bar{X})^2\right) \\ &= \mathbb{E}((X_i - \mu)^2) + 2\mathbb{E}((X_i - \mu)(\mu - \bar{X})) + \mathbb{E}((\mu - \bar{X})^2) \\ &= \sigma^2 - 2\text{Cov}(X_i, \bar{X}) + \text{Var}(\bar{X}) \\ &= \sigma^2 - 2\text{Cov}\left(X_i, \frac{1}{n} \sum_j X_j\right) + \text{Var}\left(\frac{1}{n} \sum_j X_j\right) \\ &= \sigma^2 - \frac{2}{n} \sum_j \text{Cov}(X_i, X_j) + \frac{1}{n^2} \sum_j \text{Var}(X_j) \\ &= \sigma^2 - \frac{2}{n} \text{Var}(X_i) + \frac{1}{n^2} \cdot n\sigma^2 = \frac{n-1}{n} \sigma^2. \end{aligned}$$

Variance of $\text{Bin}(n, p)$

Example

Suppose $X_1, X_2, \dots, X_n$ are independent Bernoulli random variables with $P\{X_i = 1\} = p$. Let $Y = X_1 + X_2 + \dots + X_n$. Find $\text{Var}(Y)$.

For a Bernoulli random variable,

$$\text{Var}(X_i) = p(1 - p).$$

Hence

$$\text{Var}(Y) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p).$$

Correlation coefficient

Definition

If X and Y are jointly distributed random variables and the variance and covariance of both X and Y exist and the variances are nonzero, then the correlation of X and Y , denoted by ρ , is

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

- Both numerator and denominator have the same units and so correlation has no units.
- Suppose both X and Y are subject to linear transformations, both Variance and Covariance changes but ρ does not change.
- Since ρ does not depend on the units of measurement, it is usually a more useful measure of association.

Denote by $\sigma_{XY} = \text{Cov}(X, Y)$. Then

$$\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y},$$

i.e.,

$$\sigma_{XY} = \rho \sigma_X \sigma_Y.$$

Cauchy-Schwarz Inequality

We always have $-1 \leq \rho \leq 1$. Furthermore, $\rho = \pm 1$ if and only if $P\{Y = a + bX\} = 1$ for some constants a and b .

Proof:

$$\begin{aligned} 0 &\leq \text{Var} \left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y} \right) \\ &= \text{Var} \left(\frac{X}{\sigma_X} \right) + \text{Var} \left(\frac{Y}{\sigma_Y} \right) + 2\text{Cov} \left(\frac{X}{\sigma_X}, \frac{Y}{\sigma_Y} \right) \\ &= \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} + 2 \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= 1 + 1 + 2\rho. \end{aligned}$$

Hence $\rho \geq -1$.

A similar argument shows that

$$0 \leq \text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right) = 2 - 2\rho,$$

which shows that $\rho \leq 1$.

Suppose $\rho = 1$. Then

$$\text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right) = 0,$$

which implies that

$$P \left\{ \frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} = c \right\} = 1$$

for some constant c . This is equivalent to $P\{Y = a + bX\} = 1$ for some constants a and b .

- Similar argument for $\rho = -1$.

Conditional Expectation and Prediction

We discuss the conditional mean associated to a conditional distribution.

Definition

Suppose X and Y are **discrete** random variables and that the conditional frequency function of Y given x is $p_{Y|X}(y|x)$. The **conditional expectation** of Y given $X = x$ is

$$\mathbb{E}(Y|X = x) = \sum_y y p_{Y|X}(y|x).$$

For the **continuous** case, we have

$$\mathbb{E}(Y|X = x) = \int y f_{Y|X}(y|x) dy.$$

Example

Consider a Poisson process on $[0, \infty)$ with mean λ , and let N be the number of events in $[0, 1]$. For $p \in (0, 1)$, let X be the number of events in $[0, p]$.

- Find $\mathbb{E}(X|N = n)$.
- From the assumption of a Poisson process, the count in the two intervals are independent Poisson random variables with parameters $p\lambda$ and $(1 - p)\lambda$.
- We proved before that the conditional distribution of X given $N = n$ is $\text{Bin}(n, p)$. Hence $\mathbb{E}(X|N = n) = np$.
- We also write $\mathbb{E}(X|N) = Np$. Hence $\mathbb{E}(X|N)$ is a function of N .
- In a similar way, for any two random variables X and Y , $\mathbb{E}(Y|X)$ is a random variable as a function of X .

Example

The joint density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-x/y} e^{-y} / y, & 0 < x < \infty, 0 < y < \infty; \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{E}(X|Y = y)$ (assuming that $y > 0$).

We calculated before that

$$f_{X|Y}(x|y) = \frac{1}{y} e^{-x/y},$$

that is, the conditional distribution of X given $Y = y$ is $\text{Exp}(1/y)$.

Hence $\mathbb{E}(X|Y = y) = y$.

Conditional Expectation

Just as conditional probabilities satisfy all of the properties of ordinary probabilities, so do conditional expectations satisfy the properties of ordinary expectations. For instance,

$$\mathbb{E}(g(Y)|X=x) = \begin{cases} \sum_y g(y)p_{Y|X}(y|x) & \text{in the discrete case} \\ \int_{-\infty}^{\infty} g(y)p_{Y|X}(y|x)dy & \text{in the continuous case} \end{cases}$$

and

$$\mathbb{E}\left(\sum_{i=1}^n Y_i \middle| X=x\right) = \sum_{i=1}^n \mathbb{E}(Y_i|X=x)$$

Conditional expectation given $X=x$ can be thought of as being an ordinary expectation on a **reduced sample space** consisting only of outcomes for which $X=x$.

Law of total expectation

The mean of $\mathbb{E}(Y|X)$ is given by $\mathbb{E}[\mathbb{E}(Y|X)]$.

Theorem

$$\mathbb{E}(Y) = \mathbb{E}_X[\mathbb{E}_Y(Y|X)].$$

$$\mathbb{E}(Y) = \begin{cases} \sum \mathbb{E}(Y|X=x)p_X(x) & \text{in the discrete case} \\ \int_{-\infty}^{\infty} \mathbb{E}(Y|X=x)p_X(x)dx & \text{in the continuous case} \end{cases}$$

To calculate $\mathbb{E} Y$, we may take a weighted average of the conditional expected value of Y given that $X = x$, each of the terms $\mathbb{E}(Y|X = x)$ being weighted by the probability of the event on which it is conditioned.

Proof: We only prove the discrete case. The continuous case is similar.

$$\begin{aligned}\mathbb{E}[\mathbb{E}(Y|X)] &= \sum_x \mathbb{E}(Y|X=x)p_x(x) \\ &= \sum_x \left(\sum_y y p_{Y|X}(y|x) \right) p_x(x).\end{aligned}$$

Changing the order of the summation gives

$$\mathbb{E}[\mathbb{E}(Y|X)] = \sum_y y \sum_x p_{Y|X}(y|x)p_x(x) = \sum_y y p_Y(y),$$

by the law of total probability. Therefore, $\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}(Y)$.

Example

A miner is trapped in a mine containing 3 doors, each has a tunnel behind.

The $\begin{Bmatrix} \text{first} \\ \text{second} \\ \text{third} \end{Bmatrix}$ door $\begin{Bmatrix} \text{leads to safety} \\ \text{returns to the mine} \\ \text{returns to the mine} \end{Bmatrix}$ after $\begin{Bmatrix} 3 \\ 5 \\ 7 \end{Bmatrix}$ hours of travel.

If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

Solution: Let X be the expected escape time. Then

$$\mathbb{E}(X|\text{first}) = 3, \quad \mathbb{E}(X|\text{second}) = 5 + \mathbb{E}(X), \quad \mathbb{E}(X|\text{third}) = 7 + \mathbb{E}(X)$$

On the other hand,

$$\begin{aligned}\mathbb{E}(X) &= \mathbb{E}(X|\text{first}) \cdot \frac{1}{3} + \mathbb{E}(X|\text{second}) \cdot \frac{1}{3} + \mathbb{E}(X|\text{third}) \cdot \frac{1}{3} \\ &= 5 + \frac{2}{3} \mathbb{E}(X)\end{aligned}$$

$$\Rightarrow \mathbb{E}(X) = 15$$

Example

Suppose the random variable N denotes the number of jobs in a single-server queue and X_i the service time for the i -th job. Let T denote the time to serve all jobs in the queue, i.e.,

$$T = X_1 + X_2 + \cdots + X_N.$$

Prove that $\mathbb{E}(T) = \mathbb{E}(N) \mathbb{E}(X)$. Does this agree with intuition?

By the Theorem, $\mathbb{E}(T) = \mathbb{E}[\mathbb{E}(T|N)]$.

Since $\mathbb{E}(T|N = n) = n \mathbb{E}(X)$, we see that $\mathbb{E}(T|N) = N \mathbb{E}(X)$ and so

$$\mathbb{E}(T) = \mathbb{E}[N \mathbb{E}(X)] = \mathbb{E}(N) \mathbb{E}(X).$$

This agrees with the intuitive guess that the average time to complete N jobs where N is random, is the average value of N times the average amount of time to complete a job.

Computing Probabilities by Conditioning

Let E be an event.

$$\mathbb{E}[\mathbf{1}_E] = P(E)$$

$$\mathbb{E}[\mathbf{1}_E|X = x] = P(E|X = x)$$

$$\mathbb{E}(\mathbf{1}_E) = \mathbb{E}[\mathbb{E}[\mathbf{1}_E|X = x]] = \int_{\mathbb{R}} \mathbb{E}[\mathbf{1}_E|X = x]p(x)dx,$$

which gives,

$$P(E) = \int_{-\infty}^{\infty} P(E|X = x)p(x)dx.$$

Example

Let $U \sim \text{Unif}(0, 1)$ and $X|\{U = p\} \sim \text{Bin}(n, p)$. Find the probability mass function of X .

Solution:

$$\begin{aligned} P\{X = k\} &= \int_0^1 P\{X = k|U = p\} f_U(p) dp \\ &= \int_0^1 \binom{n}{k} p^k (1-p)^{n-k} dp \\ &= \frac{n!}{k!(n-k)!} \frac{k!(n-k)!}{(n+1)!} \\ &= \frac{1}{n+1}, \quad k = 0, \dots, n. \end{aligned}$$

If a coin whose probability of coming up heads is uniformly distributed over $(0, 1)$ is flipped n times, then the number of heads occurring is equally likely to be any of the values $0, 1, \dots, n$.

Conditional variance

Recall that for a random variable Y the variance may be computed as

$$\text{Var}(Y) = \mathbb{E}(Y^2) - [\mathbb{E} Y]^2. \quad (*)$$

Now, in place of Y , we consider a conditional distribution $Y|X$.

For any value of x , we define

$$\text{Var}(Y|X = x) = \mathbb{E}(Y^2|X = x) - [\mathbb{E}(Y|X = x)]^2.$$

We may view $\text{Var}(Y|X)$ as a function of X and as a random variable.

Since

$$\text{Var}(Y|X) = \mathbb{E}(Y^2|X) - [\mathbb{E}(Y|X)]^2,$$

we have

$$\mathbb{E}[\text{Var}(Y|X)] = \mathbb{E}[\mathbb{E}(Y^2|X)] - \mathbb{E}\{[\mathbb{E}(Y|X)]^2\}.$$

and from $(*)$, $\text{Var}[\mathbb{E}(Y|X)] = \mathbb{E}\{[\mathbb{E}(Y|X)]^2\} - \{\mathbb{E}[\mathbb{E}(Y|X)]\}^2.$

Proposition

$$\text{Var}(Y) = \text{Var}[\mathbb{E}(Y|X)] + \mathbb{E}[\text{Var}(Y|X)]$$

Proof: By the law of total expectation, we see that

$$\begin{aligned}\text{Var}(Y) &= \mathbb{E}(Y^2) - [\mathbb{E} Y]^2 \\ &= \mathbb{E}[\mathbb{E}(Y^2|X)] - \{\mathbb{E}[\mathbb{E}(Y|X)]\}^2.\end{aligned}$$

Therefore, we have

$$\begin{aligned}\text{Var}(Y) &= \mathbb{E}[\mathbb{E}(Y^2|X)] - \{\mathbb{E}[\mathbb{E}(Y|X)]\}^2 \\ &= \mathbb{E}[\mathbb{E}(Y^2|X)] - \mathbb{E}\{\mathbb{E}(Y|X)^2\} + \mathbb{E}\{\mathbb{E}(Y|X)^2\} - \{\mathbb{E}[\mathbb{E}(Y|X)]\}^2 \\ &= \mathbb{E}[\text{Var}(Y|X)] + \text{Var}[\mathbb{E}(Y|X)].\end{aligned}$$

Example

Suppose that by any time t the number of people that have arrived at a train depot is a $\text{Poisson}(\lambda t)$ variable. If the initial train arrives at the depot at a time (independent of when the passengers arrive) that is uniform over $(0, T)$, what are the mean and variance of the number of passengers who enter the train?

Let $N(t)$ denote the number of passengers at time t and Y is the arrival time of the train. We are interested in $N(Y)$.

$$\mathbb{E}(N(Y)) = \mathbb{E}(\mathbb{E}(N(Y)|Y)) = \mathbb{E}(\lambda Y) = \lambda \mathbb{E}(Y) = \frac{\lambda T}{2}.$$

$$\begin{aligned}\text{Var}(N(Y)) &= \mathbb{E}(\text{Var}(N(Y)|Y)) + \text{Var}(\mathbb{E}(N(Y)|Y)) \\ &= \mathbb{E}(\lambda Y) + \text{Var}(\lambda Y) = \frac{\lambda T}{2} + \lambda^2 \frac{T^2}{12}.\end{aligned}$$

Variance of Sum of a Random Number of Variables

Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables, and let N be a nonnegative integer-valued random variable that is independent of the sequence $\{X_i\}_{i \geq 1}$.

- Find $\text{Var}(T)$, where $T = \sum_{i=1}^N X_i$.

By the earlier proposition,

$$\text{Var}(T) = \text{Var}[\mathbb{E}(T|N)] + \mathbb{E}[\text{Var}(T|N)].$$

- By $\mathbb{E}(T|N) = N\mathbb{E}(X)$, $\text{Var}[\mathbb{E}(T|N)] = (\mathbb{E} X)^2 \text{Var}(N)$.
- By $\text{Var}(T|N = n) = \text{Var}(\sum_{i=1}^n X_i) = n\text{Var}(X)$, $\text{Var}(T|N) = N\text{Var}(X)$
and $\mathbb{E}[\text{Var}(T|N)] = \mathbb{E}(N)\text{Var}(X)$.

Therefore

$$\begin{aligned}\text{Var}(T) &= \text{Var}[\mathbb{E}(T|N)] + \mathbb{E}[\text{Var}(T|N)] \\ &= (\mathbb{E} X)^2 \text{Var}(N) + \mathbb{E}(N) \text{Var}(X).\end{aligned}$$